Master's Thesis

Multi-Agent Reinforcement Learning for Joint Interference Avoidance and Gateway Load Balancing via Beam Hopping in NTN

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Abstract

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List of Abbreviations

3GPP	3rd Generation Partnership Project
ACK	Acknowledgment
BH	Beamhopping
BHO	Basic Handover
CHO	Conditional Handover
DDQN	Double DQN
DNN	Deep Neural Network
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
ECEF	Earth-centered, Earth-fixed
EIRP	Effective Isotropic Radiated Power
FLSO	Feeder Link Switchover
FSPL	Free Space Path Loss
GW	Gateway
HO	Handover
IIR	Infinite Impulse Response
KPI	Key Performance Indicator

L3	Layer 3
LEO	Low Earth Orbit
LoS	Line-of-Sight
MDP	Markov Decision Process
MTS	Minimum ToS
NLoS	Non-Line-of-Sight
NP	Non-deterministic Polynomial time
NTN	Non-Terrestrial Networks
PER	Prioritized Experience Replay
PPO	Proximal Policy Optimization
RACH	Random Access Channel
RB	Resource Block
RLF	Radio Link Failure
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
SNR	Signal-to-Noise Ratio
SINR	Signal-to-Interference-plus-Noise Ratio
TD	Temporal-Difference
TN	Terrestrial Networks
ToS	Time of Stay
UE	User Equipment
UHO	Unnecessary Handover
UFLSO	Unnecessary Feeder Link Switchover

Chapter 1. Introduction

1.1. Background and motivation

LEO satellite communication has recently emerged as a promising next-generation technology for providing global coverage. The architecture of an LEO network consists of a multitude of satellites concurrently serving numerous terrestrial cells. However, this dense multi-satellite and multi-cell environment can lead to significant inter-cell interference. Furthermore, satellites require stable and high-capacity backhaul connections with GWs to prevent data traffic bottlenecks and ensure seamless service for the cells they cover.

One of the most critical challenges in LEO satellite communications stems from the high mobility of the satellites. As satellites orbit the Earth at extremely high speeds, a ground terminal must frequently change its serving satellite. This high mobility necessitates frequent handovers and switchovers, which become a primary cause of degraded service continuity and can lead to temporary network disconnections.

In particular, conventional handover schemes based on received signal strength exhibit several limitations in this dynamic LEO environment.

Such schemes, which simply switch the connection to the satellite with the strongest signal, can trigger unnecessarily frequent handovers. Moreover, they often lead to the ping-pong effect, where handovers are rapidly repeated back and forth between two satellites at the cell edge. Consequently, the terminal's ToS with a single satellite becomes very short, which significantly degrades overall network efficiency and stability.

Therefore, to guarantee stable QoS and service continuity in LEO satellite networks, it is essential to develop intelligent handover and switchover schemes that can effectively manage the challenges posed by satellite mobility. This paper proposes a novel approach to address these pressing issues.

1.2. Contributions

To address the challenges of high mobility and dynamic network topology in LEO satellite networks, this paper proposes a novel handover and resource management scheme based on multi-agent reinforcement learning. The key contributions of this work are summarized as follows:

- **A MARL Framework for Joint Optimization:** We design a MARL framework where each satellite acts as an autonomous agent. These agents learn collaboratively to find the optimal policy for selecting both user cells and gateways simultaneously, adapting to the highly dynamic network environment without the need for a centralized controller.

- **User Link Stability by Minimizing Handovers while Maximizing SINR:** Our proposed scheme enables the satellite agents to select user cells with a sophisticated dual objective. It aims to **maximize the SINR** for all served cells while concurrently **minimizing the total number of handovers**. This approach directly mitigates the ping-pong effect and enhances service stability for ground users.

- **Feeder Link Reliability by Preventing Bottlenecks and Reducing Switchovers:** The agents intelligently select gateways to **prevent traffic bottlenecks** in the backhaul network, ensuring efficient data routing. Crucially, this selection is performed while also **minimizing the frequency of feeder link switchovers**, which enhances the stability and reliability of the satellite-to-ground connection.

The effectiveness of our proposed scheme is validated through extensive simulations, which demonstrate significant improvements in network stability and user QoS compared to conventional, signal-strength-based methods.

1.3. Overview

The remainder of this thesis is structured as follows: Sec. II discusses handover, feeder link switching, and related research in beam hopping and resource allocation. Sec. III presents the proposed multi-agent re-

inforcement learning framework in detail. Sec. IV discusses the simulation results and compares the proposed strategy with conventional approaches. Finally, Sec. V concludes the thesis.

Chapter 2. Related works

2.1. LEO satellite HO and FLSO schemes

In an NTN environment, the continuity of the service link between the UE and the satellite is ensured through HO. HO is the procedure in Fig. 2.1 by which a UE changes the serving cell. Generally, this network-controlled procedure consists of three main phases: preparation, execution, and completion. Initially, the source gNB configures the UE to perform measurements on neighboring cells, and the UE sends a MR back to the network. Based on this report and other RRM information, the source gNB decides to initiate a HO and sends a HO REQUEST to a target gNB. The target gNB performs admission control, prepares resources, and responds with a HO REQUEST ACK. The source gNB then commands the UE to switch to the target cell by sending an RCR message. Upon receiving this, the UE synchronizes with the target cell and sends an RCRC message to finalize the process. Subsequently, the target gNB requests the core network to switch the data path and triggers the release of the UE's resources at the source gNB. However, applying this procedure in NTNs presents unique challenges. Unlike in terrestrial net-

works, the difference in received signal strength between satellites is often small, making traditional measurement-based triggers less reliable. Consequently, distance- or time-based HO criteria are also employed to manage the predictable mobility of satellites. In addition to the service link, the continuity of the feeder link connecting the satellite to the ground GW is equally important. The presence or absence of a gNB onboard the satellite determines whether a feeder link switchover impacts the service link. For a transparent satellite without an onboard gNB, a feeder link switchover changes the serving GW on the ground, which in turn changes the serving gNB, resulting in a necessary HO procedure for the UE. In contrast, for a regenerative satellite equipped with an onboard gNB, a feeder link switchover can be handled at the transport layer without changing the serving gNB, making the switch transparent to the UE and avoiding an HO.

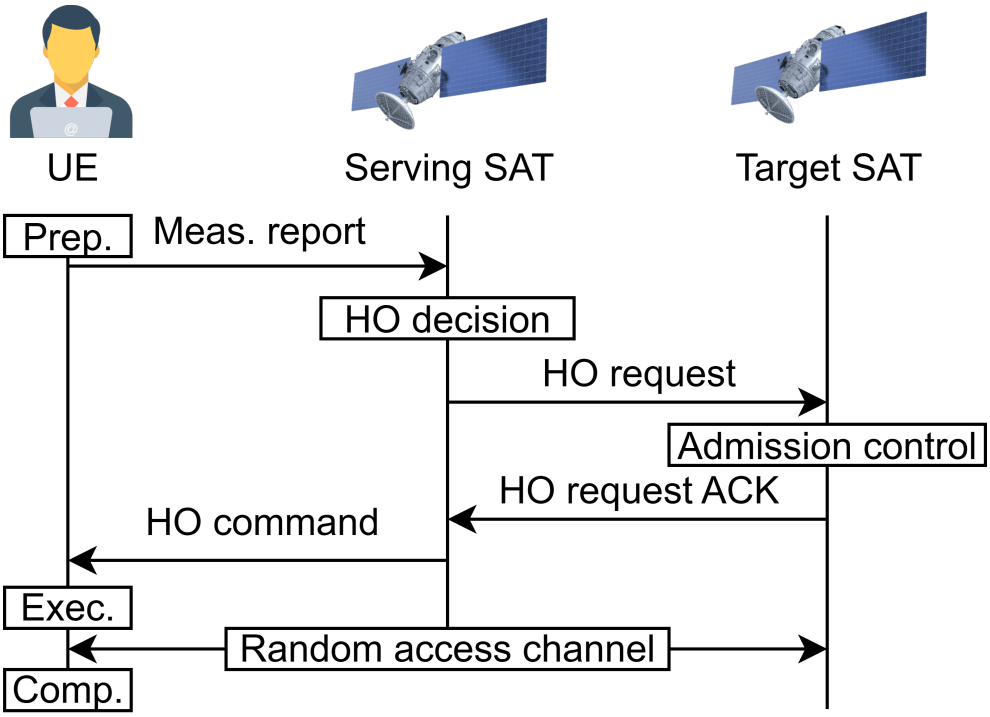


Fig. 2.1. Basic HO procedure

2.2. Reinforcement learning-based Beam hopping

BH has been extensively studied as a key resource-allocation technique to cope with spatially and temporally heterogeneous ground traffic. In LEO satellite systems, multi-satellite cooperation becomes a central challenge due to overlapping coverage and rapidly varying link conditions.

Lin *et al.* [1] proposed a framework that decomposes the multi-satellite BH problem into two core components: load balancing and interference avoidance. Exploiting the multiple-coverage property of LEO constellations, the framework first distributes hotspot traffic across several satel-

lites load balancing. Each satellite then independently designs a BH pattern for its assigned cells to satisfy per-cell demand while maintaining spatial isolation among simultaneously illuminated beams. Finally, the sequence of per-satellite BH patterns is jointly scheduled to mitigate inter-satellite interference avoidance. While this decomposition provides a clear problem structure, the approach in [1] relies on predefined heuristics to solve the subproblems sequentially, limiting adaptability to non-stationary traffic and unpredictable network dynamics.

Complementary to the above line, Xie *et al.* propose a DRL approach for multi-LEO BH with joint beam scheduling and power allocation [2]. Their method formulates a hybrid action space—discrete beam-illumination patterns and continuous per-beam power—and employs PPO with two policy heads sharing a common feature extractor. The objective jointly optimizes long-term throughput and long-term cumulative average delay under full-frequency reuse, explicitly accounting for both intra- and inter-satellite interference in the SINR model. Simulation results show notable gains over heuristic baselines (e.g., up to 8.9% higher throughput and up to 69.2% lower LTCAD), highlighting the potential of DRL for traffic-driven BH in time-varying environments. Nonetheless, the learning agent is effectively centralized and assumes access to global network state (e.g., via NOCC), and the model does not explicitly consider GW selection or feeder-link switchover effects on service continuity.

To address these limitations, this paper adopts a multi-agent reinforcement learning MARL formulation. We model each satellite as an

autonomous agent that observes local network states (e.g., per-cell demand, channel/interference conditions, and GW status) and coordinates with other agents to learn a cooperative BH policy. The learned policy simultaneously optimizes interference avoidance and gateway load balancing, thereby improving service SINR and alleviating traffic bottlenecks. In contrast to the heuristic decomposition in [1] and the centralized DRL paradigm in [2], our MARL framework natively supports distributed decision-making under partial observability and explicitly couples BH with GW selection and feeder-link switchover, which are critical for end-to-end performance in practical NTN deployments.

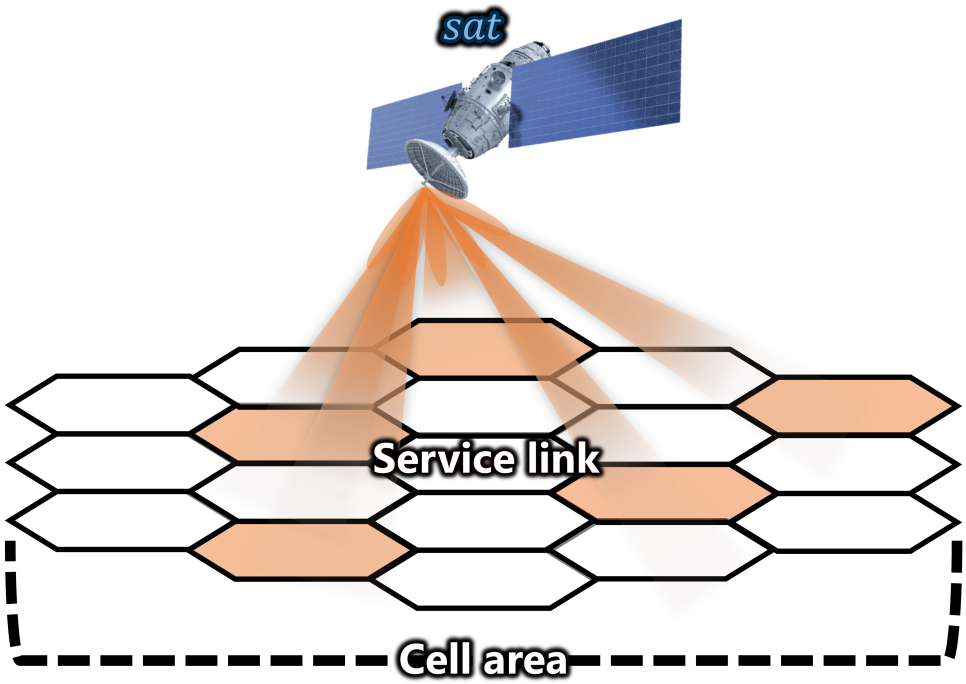


Fig. 2.2. Satellite beam hopping system

Chapter 3. Proposed multi-agent reinforcement learning-based link scheduling algorithm

3.1. System model

This thesis considers the transparent LEO satellite constellation consist of N LEO satellites, represented by the set $\mathcal{S} = \{n \mid n = 1, 2, \ldots, N\}$. The LEO satellite connects to ground GW via feeder link, where the set of GW denoted as $\mathcal{G} = \{m \mid m = 1, 2, \ldots, M\}$. The GWs consists of both an antenna and a gNB. Since the GWs are widely separated, each is equipped with its own dedicated gNB. The coordinates of each LEO satellite in the ECEF coordinate system at a given time slot t, where $t \in \{0, 1, \ldots, T\}$, are given by $l_n(t) = (x_n(t), y_n(t), z_n(t))$. As the LEO satellite services the region, it covers UEs, denoted by the set $U = \{u_t \mid u_t = N_t(\mu, \sigma^2)\}$, distributed according to a 2-D Gaussian footprint. With a hard FLSO applied, each UE within the LEO satellite's coverage area simultaneously initiates a HO to the subsequent gNB upon every FLSO. Fig. 3.1 shows the network scenario where a trans-

parent Starlink LEO satellite passes over M ground GWs in the United States, serving UEs.

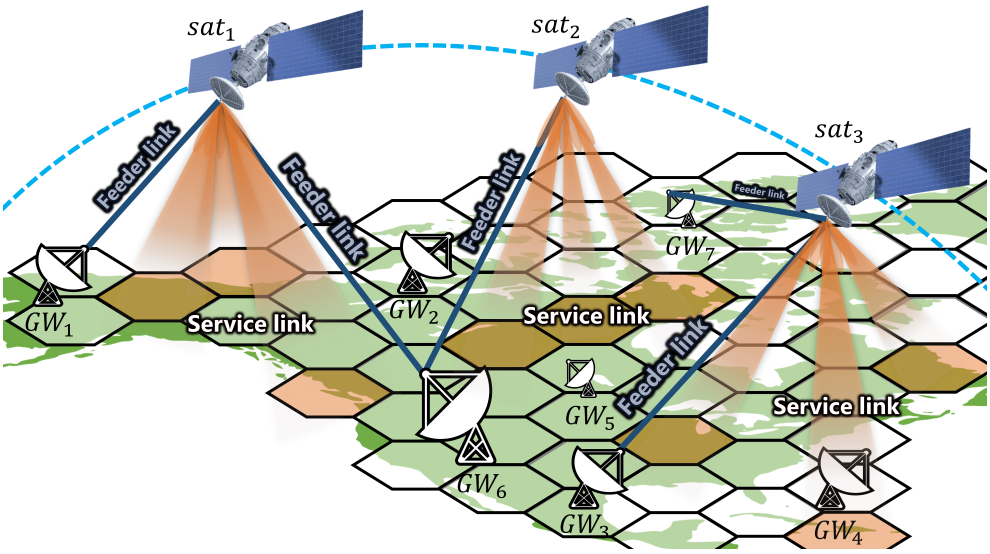


Fig. 3.1. HO and FLSO scenario: multiple satellites' beam to cell and GW

The LEO satellite n evaluates the RSRP from signals transmitted by the GW m at periodic interval T_m. The RSRP $R_{m,n}$ is calculated as follows:

$$R_{m,n} = EIRP_m + G_n - PL_{m,n} - 10\log_{10}(q_{RB}), \tag{3.1}$$

where $EIRP_m$ is the EIRP of the GW m, defined as $EIRP_m = P_m + G_m$, with P_m being the transmit power and G_m the transmit antenna gain of the GW m towards the LEO satellite n. G_n represents the receive antenna gain of the LEO satellite n. The term $10\log_{10}(q_{RB})$ is used to normalize the RSRP measurement, accounting for a single resource block out of the total q_{RB} resource blocks allocated to the link. $PL_{m,n}$ is the channel

model between the GW m and the LEO satellite n is provided in the 3GPP documentation [3] and is calculated as follows:

$$PL_{m,n} = P_{LOS,m,n}(FSPL_{m,n}(d_{m,n}, f_m) + SF_{m,n}) + P_{NLOS,m,n}(FSPL_{m,n}(d_{m,n}, f_m) + SF_{m,n} + CL_{m,n}), \quad (3.2)$$

$P_{LOS,m,n}$ and $P_{NLOS,m,n}$ denote the probabilities that the LEO satellite n to GW m is in a LOS and NLOS condition, respectively. $FSPL_{m,n}(d_{m,n}, f_m)$ denotes the FSPL as a function of the distance $d_{m,n}$ between the LEO satellite n and the GW m and the frequency f_m of the GW m uplink feeder link frequency. $SF_{m,n}$ represents the shadow fading, which accounts for the signal attenuation caused by obstacles such as buildings, trees, or terrain variations in the propagation path. $CL_{m,n}$ denotes the clutter loss, reflecting additional signal degradation due to environmental clutter. Specifically, the $FSPL_{m,n}$ is calculated as follows:

$$FSPL_{m,n} = 32.45 + 20log_{10}(f_m) + 20log_{10}(d_{m,n}). \quad (3.3)$$

At each measurement instant t, the LEO satellite n samples the RSRP from GW m and all neighboring GWs within instantaneous visibility region, yielding the raw RSRP $R_{m,n}(t)$. To mitigate the effects of shadow fading and measurement noise, these raw RSRP $R_{m,n}(t)$ samples are processed at the RRC layer using a L3 IIR filter [4]. Denoting the filtered RSRP by $\widetilde{R}_{m,n}(t)$, the update equation is calculated as follows:

$$\widetilde{R}_{m,n}(t) = aR_{m,n}(t) + (1-a)\widetilde{R}_{m,n}(t), \quad (3.4)$$

where $a \in (0,1)$ is the L3 forgetting factor, defined as $a = \left(\frac{1}{2}\right)^{K/4}$, and K is the filter coefficient. The initial condition is set as $\widetilde{R}_{m,n}(0) = R_{m,n}(0)$. Finally, the LEO satellite compiles a measurement report containing the L3-filtered RSRP $\widetilde{R}_{m,n}(t)$ of the serving and neighboring GWs and transmits this report back to the serving GW m over the feeder link.

To quantify the link quality during the simulation, we calculate the channel capacity C using the Shannon capacity formula, which is calculated as follows:

$$C = B_m \log_2(1 + \zeta), \tag{3.5}$$

where the B_m is the uplink bandwidth of GW m and ζ is the SNR calculated as follows:

$$\zeta = EIRP_m + \frac{G_n}{T} - k - PL_{m,n} - B_m, \tag{3.6}$$

where, k is the Boltzmann constant, G_n/T is the antenna gain-to-noise temperature ratio, is calculated as follows:

$$\frac{G_n}{T} = G_n - N_f - 10\log_{10}\left(T_0 + (T_a - T_0)10^{-0.1N_f}\right), \tag{3.7}$$

where N_f is LEO satellite noise figure, T_0 and T_a is ambient and antenna temperature.

3.2. Problem formulation

In this section, we address the optimization problem of performing optimal FLSO to minimize the number of HO for UE served by a LEO satellite. We formulate this problem as an NP-hard problem [5] and propose a solution by modeling it as a MDP, which is then solved using reinforcement learning techniques.

In LEO networks, frequent FLSO and HO events cause elevated latency and power consumption. FLSO $FS_{m,n}$ and HO $H_{m,n}$ are measured when the serving m transitions from $t-1$ to t calculated as follows:

$$FS_{m,n} = \sum_{t=0}^{T} \left(1 \text{ if } m_t \neq m_{t-1} \text{ else } 0\right), \tag{3.8}$$

$$H_{m,n} = \sum_{t=0}^{T} \left(u_{t-1} \text{ if } m_t \neq m_{t-1} \text{ else } 0\right). \tag{3.9}$$

To address these issues, we formulate an optimization problem that enables the LEO satellite to perform optimal FLSO while minimizing UE HO and UFLSO, and maximizing ToS and RSRP, meeting the conditions

calculated as follows:

$$\textbf{P1:} \quad \min_{m} \sum_{n=1}^{N} \sum_{m=1}^{M} \sum_{t=0}^{T} \left(FS_{m,n} + H_{m,n} + UFS_{m,n} - ToS_{m,n} - \widetilde{R}_{m,n} \right), \tag{3.10}$$

s.t.

$$\textbf{C1}: \quad \theta_{m,n} \geq \theta_{\min}, \quad \forall n, \forall m, \forall t, \tag{3.11}$$

$$\textbf{C2}: \quad \widetilde{R}_{m,n} \geq R_{\min}, \quad \forall n, \forall m, \forall t, \tag{3.12}$$

$$\textbf{C3}: \quad I_{m,n}(t) = 1, \quad \forall n, \forall m, \forall t. \tag{3.13}$$

In **P1**, $UFS_{m,n}$ represents the cost of an UFLSO, which occurs when the LEO satellite n switches to another GW within the MTS and $ToS_{m,n}$ represents ToS the duration of the LEO satellite n remains connected to the GW m. The objective function minimizes the total cost, which includes the FLSO, HO of UEs and UFLSO events and maximizes ToS and RSRP across N LEO satellites, M GWs, and T time slots. Constraint **C1** ensures the elevation angle $\theta_{m,n}$ between LEO satellite n and GW m exceeds the minimum threshold $\theta_{\min}$, guaranteeing reliable communication by maintaining a sufficient line-of-sight. Constraint **C2** ensures the filtered RSRP $\widetilde{R}_{m,n}$ exceeds the minimum required rate $R_{\min}$, ensuring adequate link quality for effective data transmission. Constraint **C3** enforces that each LEO satellite connects to exactly one GW at all time t, using the indicator variable $I_{m,n}(t)$, which is 1 if LEO satellite n is connected to GW m at time t.

3.3. Proposed algorithm

To address the problem, we transform it into a dynamic programming framework to derive an optimal FLSO policy. We model the FLSO optimization problem as an MDP due to its ability to handle sequential decision-making in dynamic environments.

3.3.1 MDP modeling

The LEO satellite's state s_t evolves over time and actions a_t impact future reward r_t with discount factor γ [6]. This framework enables the LEO satellite to learn an optimal FLSO policy through reinforcement learning, effectively balancing service continuity and HO minimization. The state space s_t of the MDP is calculated as follows:

$$s_t = \left\{ l_n(t), m_t, \widetilde{R}_{m,n}(t), u_t \right\}. \tag{3.14}$$

The state is defined as a tuple comprising the LEO satellite's ECEF coordinates $l_n(t)$, the currently connected GW m_t, the RSRP of the current GW $\widetilde{R}_{m,n}(t)$, and the number of UEs currently being served u_t at time t, configured to be accessible and interpretable by the LEO satellite.

The action space a_t involves the LEO satellite selecting the optimal GW at each step. Accordingly, our action space is calculated as follows:

$$a_t = \{m_t | m_t \in \mathcal{G}_t\}. \tag{3.15}$$

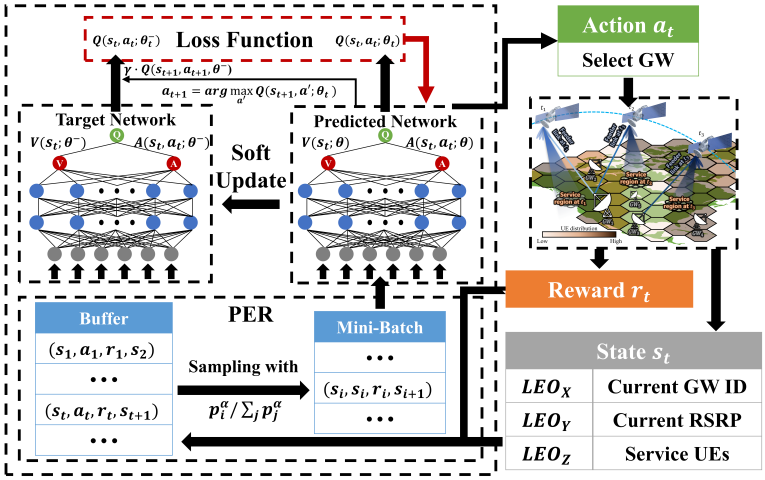


Fig. 3.2. The architecture of the proposed dueling DQN for FLSO optimization in LEO satellite networks.

The action comprises the set of GW $\mathcal{G}_t$ denotes the set of all available GW at t.

The immediate reward r_t is formulated to jointly promote service continuity and signal quality while penalizing both necessary and UFLSO. It is calculated as follows:

$$r_t = (1 + \rho u_t)FS(t) + UFS(t) - \Big(ToS(t) + \nu \widetilde{R}_{m,n}(t)\Big). \tag{3.16}$$

The parameters ρ and ν are normalization factors used to align the values with other metrics.

Discount factor γ is a value between 0 and 1, which represents the rate at which future rewards are discounted when considering them at the current step.

3.3.2 Proposed dueling DQN algorithm

In this paper, we employ a reinforcement learning algorithm to enable LEO satellites to operate in an dynamic environment. As depicted in Fig. 2, the architecture of the proposed dueling DQN algorithm, incorporating a ϵ-greedy policy, is designed to optimize FLSO in LEO satellite networks. The LEO satellite receives feedback to execute actions that sustain both a prolonged ToS and an elevated RSRP, while minimizing the frequency of FLSO, UFLSO, and UE HO events. We enhance the traditional DQN framework by incorporating DDQN, PER, and dueling DQN to improve stability, learning efficiency, and robustness against noise in LEO satellite communication systems.

The DQN algorithm [7] approximates the Q-function $Q(s_t, a_t; \theta)$, which maps a state s_t and action a_t to an expected cumulative reward, using a DNN parameterized by θ. Two DNNs are utilized: an online network for action selection and a target network for value estimation. The Q-function is trained to minimize the following loss function and loss function is calculated as follows:

To improve robustness against random noise, such as channel shadow fading and variations in UE distribution, we utilize a dueling DQN [8]. The dueling DQN architecture enhances the traditional DQN by splitting the Q-function into two separate streams: the state-value function $V(s; \theta)$ and the advantage function $A(s, a; \theta)$. The Q-function is then calculated

Algorithm 3.1 Dueling DDQN-based Algorithm for UE HO

Initialization: Random weights θ, replay memory $\mathcal{R}$ with capacity $\mathcal{D}$, minibatch size $\mathcal{Z}$, total episodes $\mathcal{E}$, prioritization parameter α, importance sampling parameter β, exploration rate ϵ, Polyak averaging parameter τ
Input: State, action, discount factor
Output: Optimal FLSO decision
for *Episode* $= 1$ to $\mathcal{E}$ **do**
    Initialize the integrated environment
    for $t = 0$ to T **do**
        $a_t = \begin{cases} \arg\max_{a'} Q(s_t, a'; \theta) & \text{with } 1 - \epsilon \\ extrandomaction & \text{with } \epsilon \end{cases}$
        Execute action, observe reward and next state
        Store sample (s_t, a_t, r_t, s_{t+1}) into $\mathcal{R}$
        if $\mathcal{R} > \mathcal{D}$ **then**
            for $i = 1$ to $\mathcal{Z}$ **do**
                Sample (s_i, a_i, r_i, s_{i+1}) from $\mathcal{R}$ with probability p_i^α
                Calculate target network
            end for
            $\theta^- \leftarrow \tau \cdot \theta + (1 - \tau) \cdot \theta^-$
        end if
    end for
end for

as follows:

$$Q(s,a;\theta) = V(s;\theta) + \left(A(s,a;\theta) - \frac{1}{|A|} \sum_{a'} A(s,a';\theta) \right), \tag{3.17}$$

where $V(s;\theta)$ represents the value of the state s, $A(s,a;\theta)$ represents the advantage of taking action a in state s, and $|A|$ is the number of possible actions. This decomposition allows the network to better distinguish the value of a state from the advantage of each action, improving robustness against noise and enhancing learning efficiency in complex LEO satellite environments.

To further enhance the learning efficiency, we incorporate PER as proposed in [9]. PER prioritizes experiences with higher TD errors, where the TD error δ_i for transition i is defined as the difference between the target and current Q-values. The priority is calculated as follows:

$$p_i = |\delta_i| + \epsilon_p, \tag{3.18}$$

where $\delta_i = r_i + \gamma Q(s_{i+1}, a_{i+1}; \theta^-) - Q(s_i, a_i; \theta)$ is the TD error, and ϵ_p ensures non-zero priorities. PER uses $\alpha \in [0,1]$ to control sampling probability p_i^α and $\beta \in [0,1]$ to adjust importance sampling weights $(\mathcal{D} \cdot P(i))^{-\beta}$, where $\mathcal{D}$ is replay buffer limit and $P(i)$ is the probability of sampling transition i. This approach accelerates learning by focusing on significant transitions, which is beneficial in LEO satellite networks with dynamic and noisy conditions.

By combining dueling DQN with DDQN and PER, our approach

effectively addresses overestimation bias, accelerates learning, and mitigates the effects of random noise, leading to improved LEO satellite HO performance with prolonged ToS, elevated RSRP, and reduced FLSO, UFLSO, and UE HO events. The detailed process of the training is described in Algorithm 1.

Chapter 4. Performance evaluation

4.1. Simulation setup

4.2. Key Performance Indicators

4.3. Results and Discussion

Chapter 5. Conclusion

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